

WEB-BASED ML MODEL EVALUATOR

Use Case Flow

UC-01 Register

1.2 Analyst Registration

Primary Actor: Analyst

Main Flow:

- 1 The Analyst requests the registration page.
- 2 The Analyst provides necessary registration information.
- 3 The system validates the provided information.
- 4 The system checks that no account using the specified details exists.
- 5 The system creates the Analyst account.
- 6 The system informs the Analyst about the successful creation and opens the Analyst dashboard.

Alternative Flow:

If details provided are invalid and/or incomplete, an error message will be displayed asking the Analyst for a correction.

If an account using the specified details already exists, the registration will not be successful. An error message stating this will be presented.

A registration never creates an Admin account; an Admin account can only be granted later by an already existing Admin (see Manage User Roles)

UC-02 Login

Analyst / Admin (Primary Actor)

Main flow:

The user navigates to login.

The user fills in their credentials.

The system validates their credentials.

The system authenticates the user.

The system navigates the user to the role specific dashboard (Analyst/Admin).

Alternative Flow:

If their login details are incorrect the user is presented with an error and is again prompted to enter their details.

If the users account is blocked the system denies login.

UC-03 Upload Dataset

Primary Actor: Analyst

Main Flow:

1. Analyst opens the dataset upload screen.
2. Analyst choose the file(s) containing the dataset(s).
3. Analyst chose the target column for the dataset.
4. System validates file format & schema; acceptance of the dataset file.
5. System loads the data so Analyst can evaluate it.

Alternative Flow:

The system rejects the file if corrupt/invalid: The system gives a readable explanation of what's wrong (e.g. "Cannot load-file is corrupt").

UC-04 Upload Model

Subject: Analyst

Main Sequence of Events:

Analyst access models upload function.

Analyst selects trained models file.

File format validation takes place.

Model is made available.

If dataset uploaded previously, check its expected feature set to be compatible with its schema.

Extensions:

Invalid model file or an inappropriate model format are immediately rejected by the system.

Incompatible with the previous datasets model, reject, explain.

UC-05 Select Task Type

Actor: Analyst.

Main flow.

The Analyst opens task selection part.

The Analyst selects classification or regression.

The system verifies the selection with uploaded data set and model.

The system activates the appropriate and available option for metric calculation and visualizations corresponding to the selected task.

Alternative flow:

If the selected task is incorrect, for example, classification model with a regression dataset/ vice-versa, system shall deny to perform evaluation and show an indication or reasoning.

UC-06 Evaluate Model

Primary Actor: Analyst

Main flow:

The analyst goes to evaluation screen of uploaded model/dataset pair.

The system revalidates that the model and dataset can be used together.

The system run model against the evaluation data-set and gets the prediction result.

The system calculate the metrics relevant to the stated task-type.

The system render the visualization appropriate to the task.

The system record the evaluation-the result of evaluation (metric-values) and visualization, alongside with the metadata like-model ID, dataset ID, task type and the time-against the Analysts account.

The system shows the result to the analyst.

Alternative flow:

If any exception is thrown during prediction or calculation of metrics, system terminates the evaluation without storing any partial evaluation result and display particular message which tell the reason why it terminates, Analyst will try by fixing the error(s).

UC-07 View Metrics

Actor: Analyst

Main: Analyst opens completed analysis

(a) System will show metric values on that task.

Alternative: The analysis is not available: System will show a suitable message.

UC-08 View Visualizations

Actor Principal: Analyst

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Analyst opens an evaluation with execution finalized.

System plots the visualizations defined in task (confusion matrix and ROC curve for classification tasks, if probabilities were predicted and R graph for regression tasks).

FluxAlternativo:

If the model did not predicted the probabilities for the prediction it should't plotROC curve and plot only the possible visualizations that has been computed.

UC-09 Compare Models

Actor: Analyst

Main Flow:

The Analyst opens the comparison.

The Analyst selects two or more completed evaluations which are of the same task type.

Their respective metrics are presented side-by-side on the screen.

Alternate Flow:

If selected evaluations were not of same type, system warns of this condition and doesn't enable comparison.

UC-10 View Evaluation History

Primary Actor: Analyst

Main Flow:

The Analyst access the evaluation history page.

The system populates the evaluation history for the analyst (i.e. The analyst saved evaluations).

The analyst selects an evaluation that the analyst want to reopen it with.

Alternative Flow:

If no saved evaluation history, the system populates "no evaluation history found".

UC-11 Generate Report

Primary Actor: Analyst

Main Flow:

- Analyst chooses one saved evaluation or two or more saved evaluations for comparison.
- System aggregates model, dataset, metrics, and visualization information for each choice.
- System outputs aggregated information as a report.

Alternative Flow:

- System does not retrieve and aggregate information on one or both of the saved evaluations due to it no longer existing, displaying an error to the user and not generating the report.

UC-12 Manage User Accounts

User(s): Admin

Main Scenario:

The Admin navigates to the manage account section.

A list of registered accounts with their statuses is shown by the system.

The Admin selecting an account, Suspend it or Reactivate It.

The System verified that the admin will not manipulate their personal account and after the said operation the system still have at least an active admin.

The account status is updated by the System.

The System add the activity on audit log (AdminID, effected UserID andtimestamp).

Exception Scenario:

The system does not affect the activity and informs the admin if the effected user already having that specified state.

If the system finds that Admin will manipulate personal account or after will have no remaining active administrator for system, system refuse and informed the user.

UC-13 Manage User Roles

Subject: Userrole Management

Primary Actor: Admin

Main Flow:

- The admin enters the role management part.
- The system shows all existing accounts and their corresponding roles.
- The admin chooses an account and gives them the Analyst role or Admin role.
- The admin confirms the new entry.
- The system updates the new account with its new role.
- The system log the new entries using the Admin ID, the affected account ID, the old role, the new role, and timestamp.

Alternate Flow:

- Admin cannot modify its role itself
- System denies the request if it downgrades to the last Admin account.